

SafeConf | review packet

Public perturbation evidence enables cold-start risk auditing for single-cell perturbation prediction

投稿目标：IEEE/ACM Transactions on Computational Biology and Bioinformatics（TCBB，CCF-B方向）

证据冻结版本：2026-10-09；当前Git提交：648e5b1；模型训练和数据下载均未由本PDF构建触发。

本文件用于研究组审阅和投稿前校对，正文的最终期刊排版仍由LaTeX模板完成。

Abstract

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Single-cell perturbation predictors are usually evaluated after the experiment has been performed, while practical use requires deciding which predictions deserve limited experimental review. Existing post-hoc reliability methods learn from the target predictor's own out-of-fold errors. This creates a cold-start problem: a new predictor has no target-screen error history when its first predictions need to be audited. We introduce SafeConf, a risk-auditing framework that uses protocol-compatible public perturbation experiments as an external evidence layer and adds predictor-specific feedback only when it is available. The public layer produces a frozen risk ranking without fitting target-screen error labels; source-model errors and target feedback are tracked as separate information budgets and are not forced into the default system.

On a fixed McFaline benchmark, public experimental evidence identified 22 and 24 of the 43 most erroneous predictions for two different predictors, compared with 4 and 5 using prediction magnitude alone. The remaining mean error after the same review budget was reduced by 7.07% and 6.84%. In a retrospective Frangieh cross-family stress test, public evidence improved risk utility over magnitude in both GEARS-to-scGPT and scGPT-to-GEARS directions, with paired U20 increments of 0.460 and 0.309 respectively. A frozen KOLF study supplied an independent evaluation contract with 600 upstream-training, 300 development, and 300 confirmation genes. One predictor passed the pre-registered competence gate; the public rule covered 228 of 300 evaluation tasks and used a deterministic magnitude fallback for the remaining tasks. The external point estimate favored public evidence over magnitude, while the gene-cluster interval remained wide. These results support public perturbation history as a practical cold-start risk layer and define the conditions under which predictor-specific adaptation should be used.

Study design and adopted method

SafeConf将公共真实扰动实验、Source错误和Target反馈登记为三类信息。默认系统只使用PublicRule；Source和Target作为有预算、有边界的候选扩展。

McFaline：212任务/152基因簇；KOLF：600上游训练、300开发、300确认基因。公共历史不读取当前确认任务答案；风险分数在确认读取前冻结。

KOLF共有公共响应轴为1366/1400。缺失坐标不补零；公共参照只在实际共同轴上计算。无历史任务走固定幅度回退。

主端点是20%复核的U20、严重错误命中和复核后的剩余误差；区间按扰动基因簇进行5000次配对bootstrap。

Figure 1 | Zero target-error risk auditing

Fixed20% review: observed severe-error retrieval

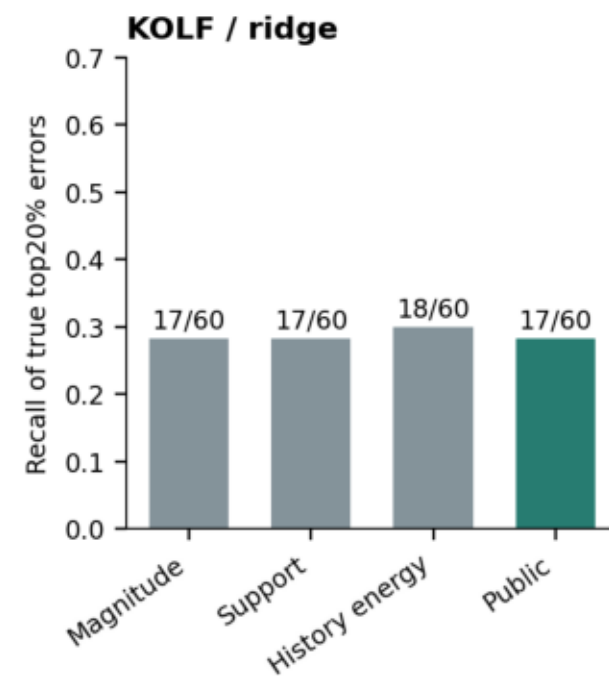
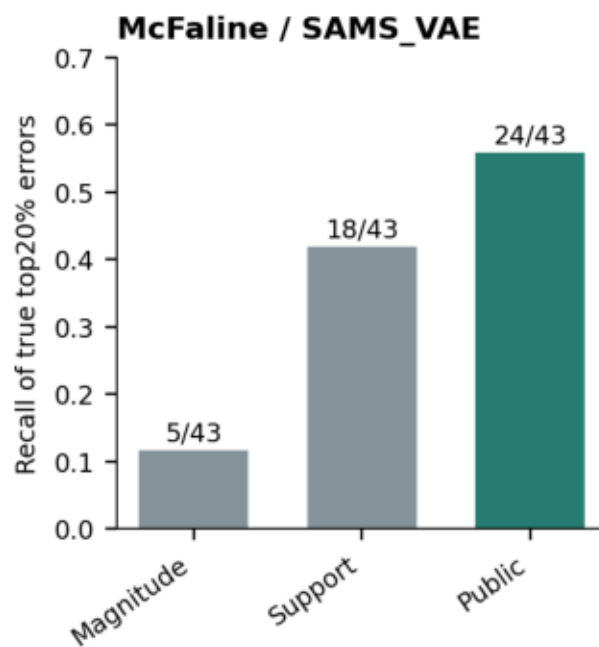
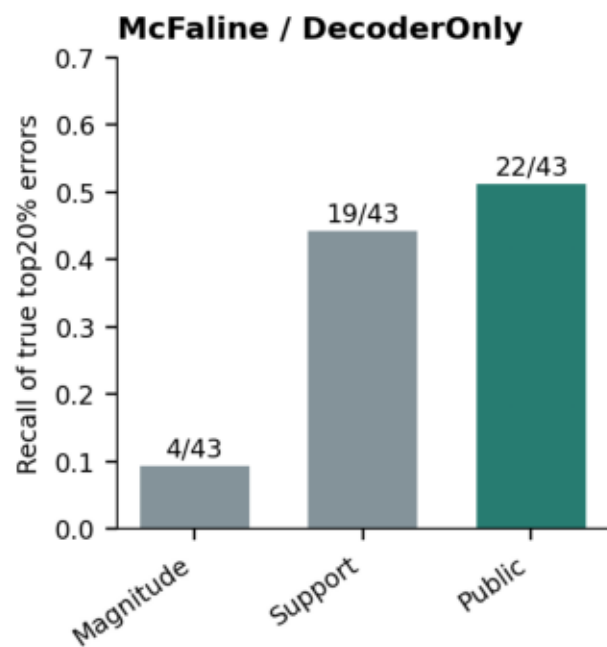


Figure 2 | Content attribution

20 support-matched nulls; strict content increment not established

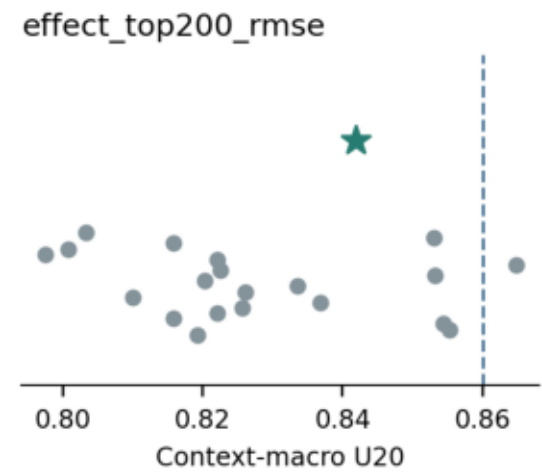
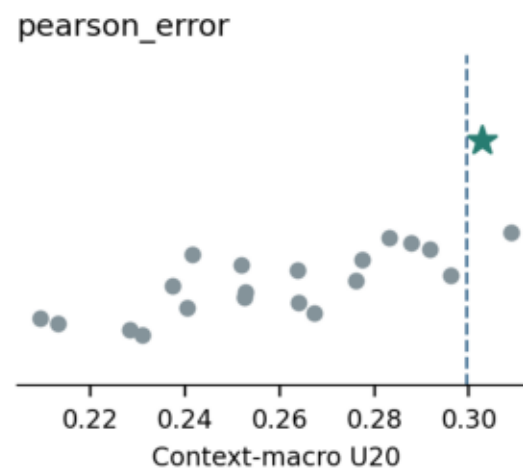
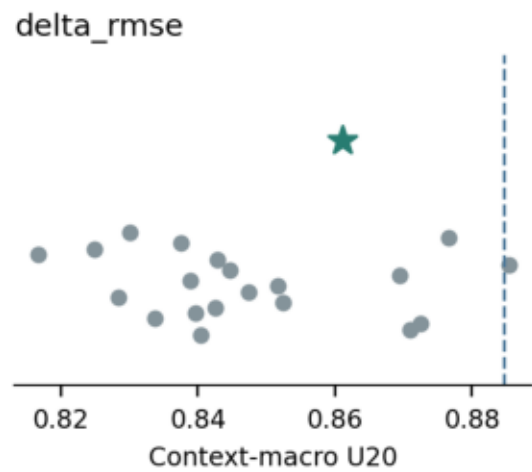
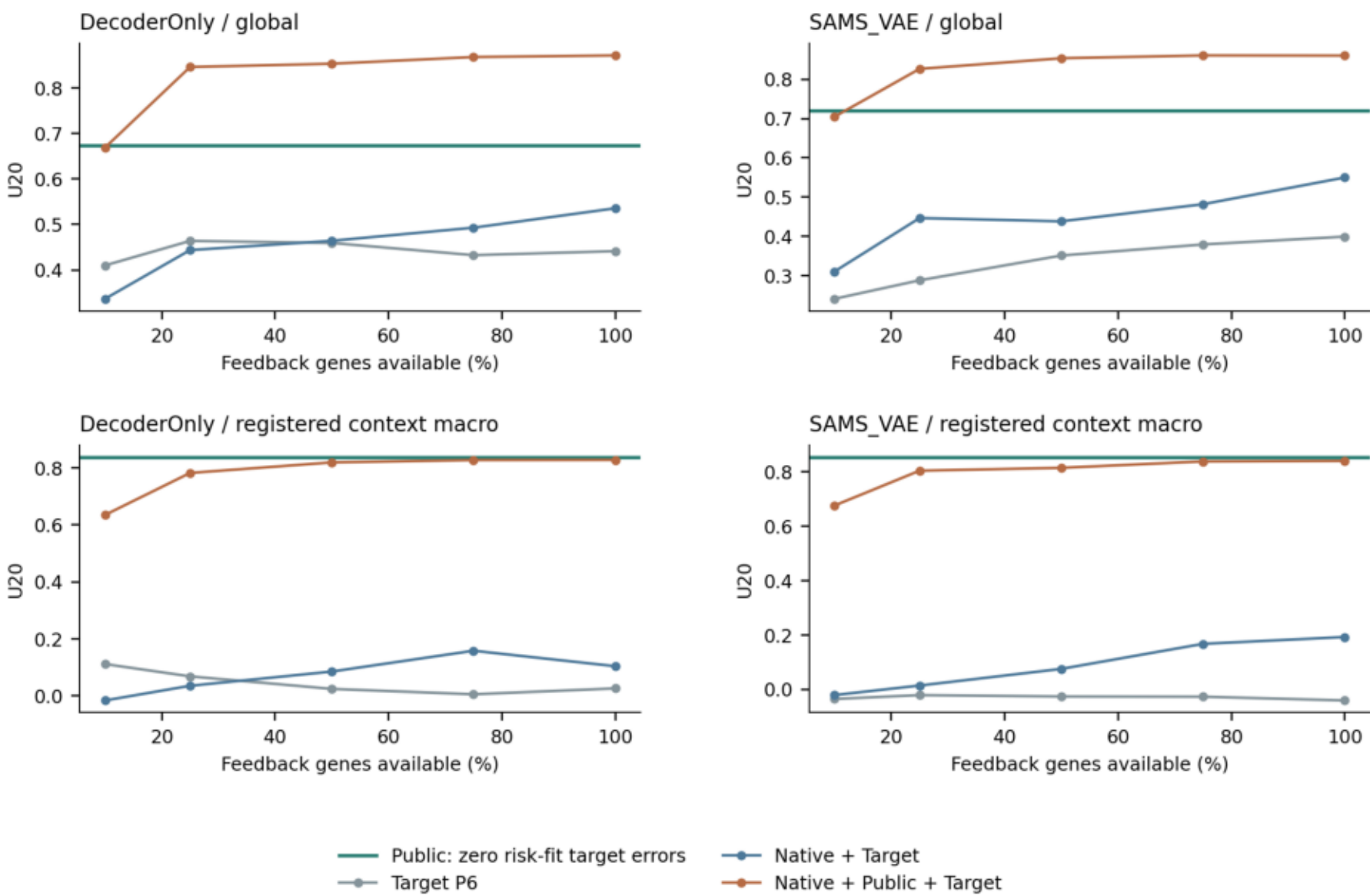


Figure 3 | Feedback budget
McFaline SEEN evidence: global and macro outcomes stay separate



McFaline全量global和context-macro端点分开展示。

Figure 4 | KOLF frozen independent evaluation

KOLF: point gains, broad intervals; no extra severe-error hits

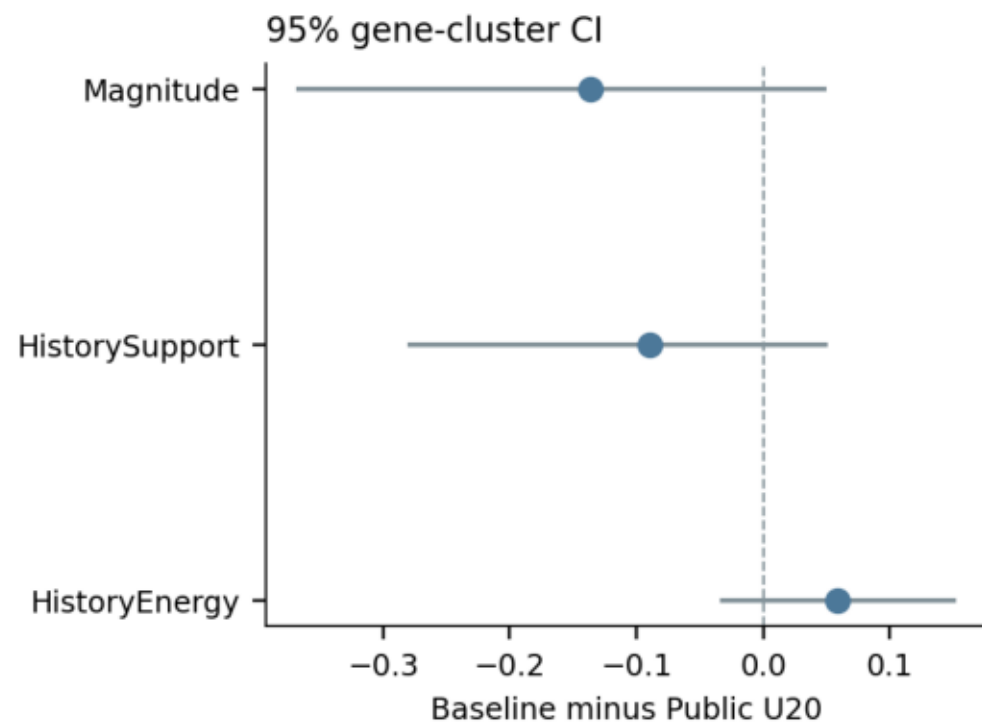
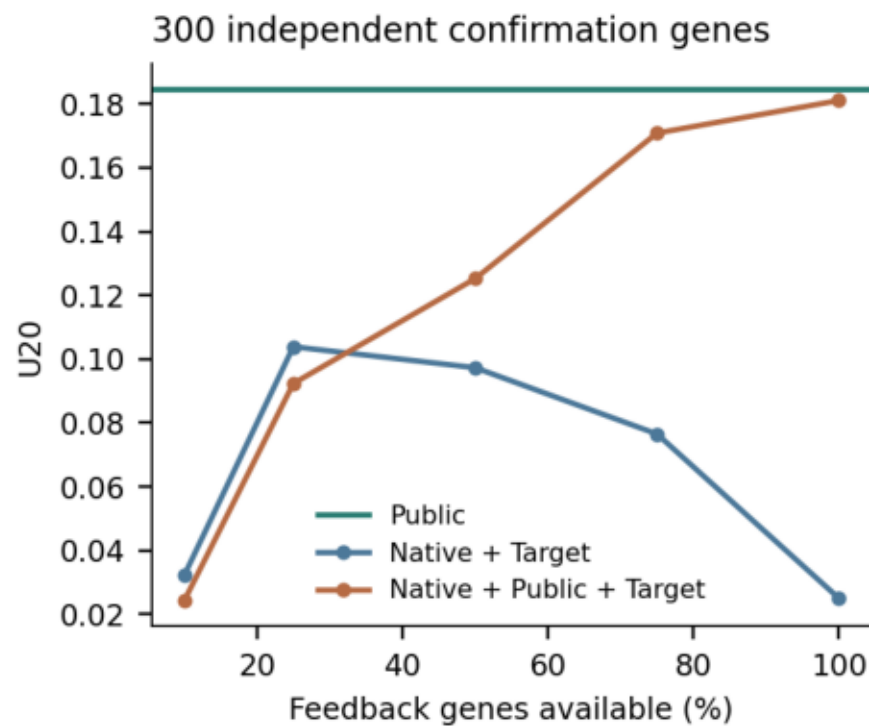
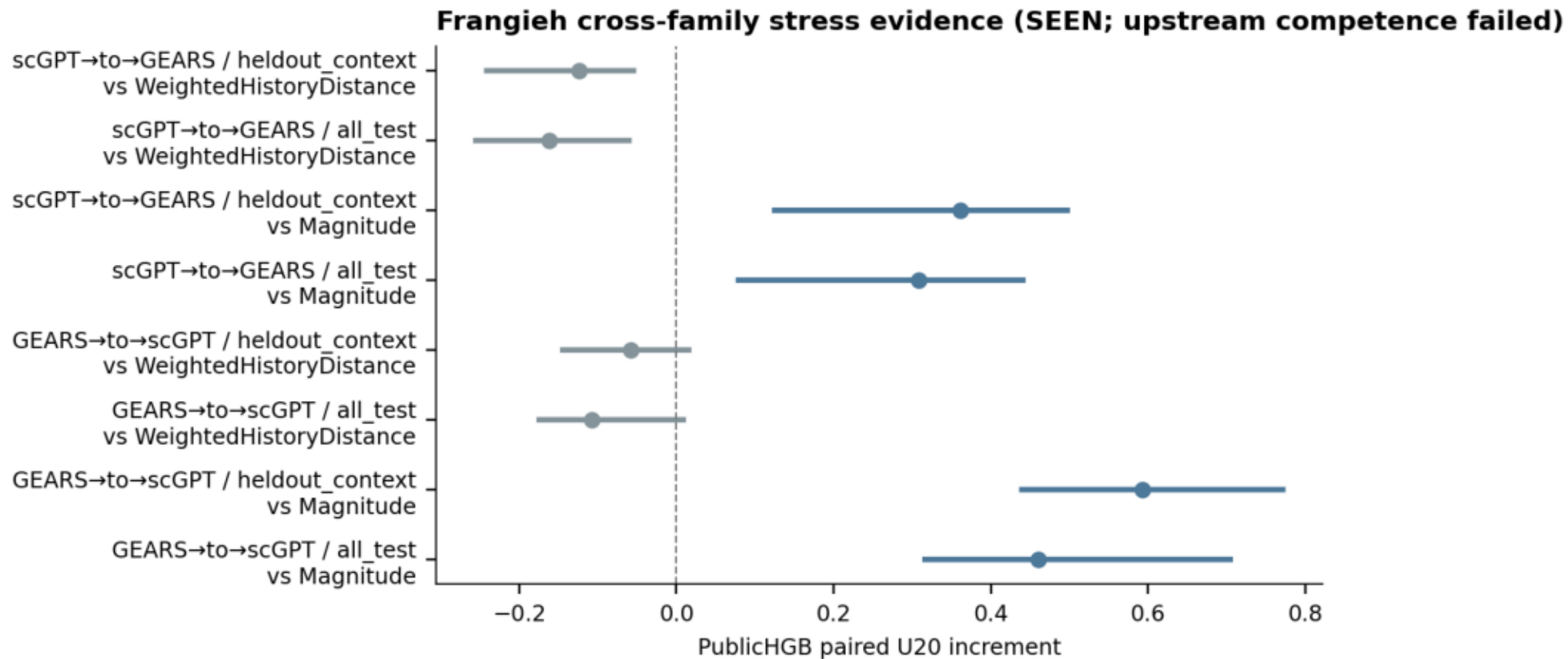


Figure 5 | Cross-family stress evidence



GEARS↔scGPT压力证据；六个上游能力门失败，不能当作合格独立确认。

Key results and claim boundaries

McFaline：Public在DecoderOnly和SAMS上分别命中22/43与24/43个最高误差任务，幅度基线为4/43与5/43；剩余误差下降7.07%与6.84%。

Frangieh：PublicHGB相对Magnitude的paired U20增量为+0.460 [0.317,0.704]与+0.309 [0.079,0.440]；上游能力全部失败，保留为stress evidence。

KOLF：Ridge通过能力门；Public覆盖228/300；Public相对Magnitude点增量+0.136，95%区间跨零；严重错误命中17对17，因此解读为可运行性和信息账本验证，而非已确认普遍优越。

当前论文主张是公共证据可启动风险审核和跨模型压力下仍有价值；不声称所有学习器、所有背景、所有反馈预算均单调改善。